

Data Science for Software Engineering

Assignment 3 Report

Hadoop YARN - Exploring Software Design Issues Using
LDA and BERTopic



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Group 10

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1 Introduction

Apache Hadoop YARN (Yet Another Resource Negotiator) serves as the core resource management framework of the Hadoop ecosystem, responsible for scheduling applications and managing computational resources across distributed clusters. As an actively maintained open-source project, Hadoop YARN continuously evolves through discussions recorded in its Jira issue tracking system. These issues document software design decisions, feature requests, bug reports, architectural enhancements, and implementation challenges, thereby providing a valuable source of information for understanding the evolution of the software system. However, the large volume of textual issue reports makes it difficult to manually identify recurring design concerns and the underlying themes discussed by developers.

The main objective of this study is to analyze the discussions recorded in the Hadoop YARN Jira issue repository in a systematic manner. Topic modeling techniques provide an effective way to identify common discussion topics from a large collection of issue reports. In this study, LDA and BERTopic are used to identify the main discussion topics in the Hadoop YARN issue repository. In addition, ontology-based preprocessing is applied to improve the quality of the generated topics. The identified topics are then analyzed to study their characteristics and their relationships with software design decision types. This study aims to provide a better understanding of the software design discussions within Hadoop YARN and evaluate the effectiveness of topic modeling techniques for software engineering research.

1.1 Background

1.1.1 Issue Tracking Systems and Topic Modeling

Issue tracking systems play an important role in modern software development by recording bug reports, feature requests, design discussions, and implementation decisions throughout the software lifecycle. Projects such as Apache Hadoop YARN use Jira to manage these discussions, creating a large collection of textual issue reports. These reports contain valuable information about software evolution and design decisions. However, manually analyzing thousands of issue reports is difficult and time-consuming, making it challenging to identify common discussion topics and recurring software engineering concerns.

Topic modeling is a text mining technique that automatically discovers common topics from a collection of documents. Instead of reading each issue individually, topic modeling groups issues that discuss similar concepts based on the words they contain. This allows researchers to obtain an overview of the main themes discussed within a software project and better understand the evolution of software design.

1.1.2 LDA and BERTopic for Software Engineering

This study employs two topic modeling techniques: Latent Dirichlet Allocation (LDA) and BERTopic. LDA is a probabilistic topic modeling algorithm that identifies topics by analyzing the frequency of words across documents. In contrast, BERTopic uses transformer-based language embeddings to capture the contextual meaning of words, allowing it to generate more semantically meaningful topics.

To improve the quality of the identified topics, this study also applies ontology-based preprocessing by replacing selected software engineering terms with predefined ontology classes before topic modeling. After generating the topics, statistical analyses are performed to examine their characteristics and identify significant relationships among topics and software design decision types. These analyses provide a comprehensive understanding of the major software engineering discussions within the Hadoop YARN issue repository.

The following section presents the study design and methodology to investigate these research questions.

2 Study Design

The study aims to address the following questions regarding the intersection of structural analysis and artificial intelligence:

- **RQ1:** What topics emerge from LDA and BERTopic and what are common keywords per topic?
- **RQ2:** What are the characteristics of the issues in each topic?
- **RQ3:** What topics significantly co-occurrence together among LDA topics and BERTopics? What topics from LDA and BERTopics significantly co-occur with the types of design decisions?

The results obtained from applying these methods are presented in the next section.

3 Results

The results obtained from the applied methods provide an initial understanding of the structural patterns within the Hadoop YARN Capacity Scheduler component, focusing on how different algorithms group the classes and the structural similarity of their recovered architectures.

3.1 RQ1: Design Topic Discovery

3.2 RQ2: Topic Characterization

3.3 RQ3: Topic Co-occurrence Analysis

The following section discusses these findings in greater detail, interpreting their implications for architectural recovery.

4 Discussion

To address the first research question, we interpret the observed clustering outcomes in relation to architectural accuracy and component identification. The following discussion examines how effectively the applied algorithms capture meaningful structural boundaries within the Hadoop Yarn system.

4.1 RQ1: Design Topic Discovery

4.2 RQ2: Topic Characterization

4.3 RQ3: Topic Co-occurrence Analysis

The final section summarizes our key findings and outlines directions for future work.

5 Threats to Validity

Although our results provide a solid baseline, it is important to acknowledge the limitations and potential sources of errors that can affect the validity of our findings. These are categorized into three areas:

5.1 Construct Validity

5.2 External Validity

5.3 Reliability

6 Conclusion and Future Work

References

7 Appendix

7.1 Teammates' Contribution

Teammate	Week 1
Mihir Sane	4h
Aryan Solanki	4h
Sujit Maniya	4h
Jenish Jai	4h
Indudhara S	4h

Table 1: Teammates' contribution summary

7.2 Shell Scripts